

Sizing Systems Engineering Activities to Optimize Return on Investment

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Abstract. This work continues significant results from the Systems Engineering Return on Investment (SE-ROI) project that has quantified systems engineering (SE) success using extensive interviews with real programs. In prior work the project reported that effective systems engineering (SE) at a level of approximately 15% of a project cost optimizes cost, schedule, and stakeholder success measures. The project also identified a set of effective characterization parameters for quantifying SE. This work explores the relationship between the characterization parameters and the success correlations. It is shown that eight major SE activities can be “right-sized” based on the characterization parameters of a program as an *a priori* estimation method to optimize the ROI of the activities. The paper shows the quantified relationships, based on the database of interviews. Two example programs demonstrate the method and illustrate the typical optimized level of SE activities.

Background

Prior work on the Systems Engineering Return on Investment (SE-ROI) project has quantified systems engineering (SE) success based on both surveys and extensive interviews. The work has progressed over a period of 12 years with major results.

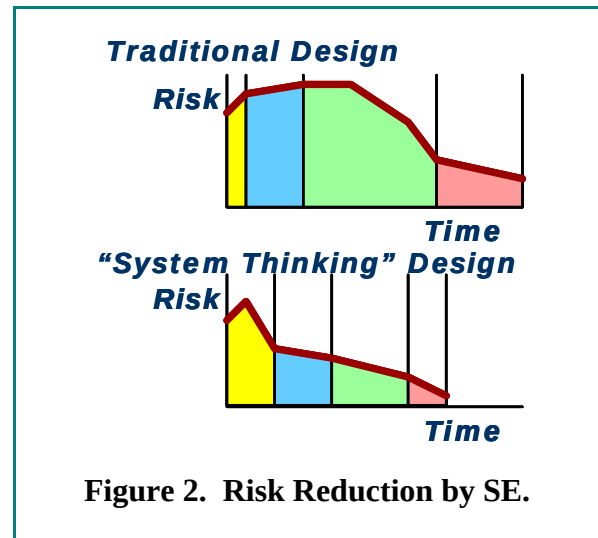
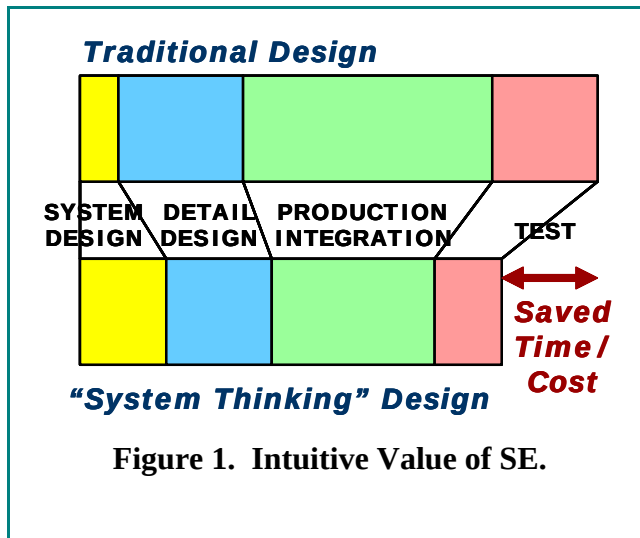
The challenges of developing and sustaining large complex engineering systems have grown significantly in the last decades. The practices of systems engineering promise to provide better systems in less time and cost with less risk, and this promise is widely accepted in some industries. However, we have long lacked specific evidence regarding the right amount of systems engineering to bring about the best results, as well as the correct timing for the application of system engineering.

The intuitive understanding of the value of SE is shown in Figure 1. In traditional design, without consideration of SE concepts, the creation of a system product is focused on production, integration, and test. In a “system thinking” design, greater emphasis on the system design creates easier, more rapid integration and test. The overall result is a savings in both time and cost, with a higher quality system product.

The primary impact of the systems engineering concepts appears to be to reduce risk early, as shown in Figure 2. By reducing risk early, the problems of integration and test are prevented from occurring, thereby reducing cost and shortening schedule. The challenge in understanding the value of SE is to quantify these intuitive understandings.

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Theoretical relationships were explored by the author concerning optimizing systems engineering value (Honour 2001) and quantitative relationships (Honour 2002). One key result was the core hypothesis of the current work, which is that there exists a quantitative relationship between the systems engineering activities and the success of the program. In the Value of Systems Engineering work (Honour 2004), the author quantitatively demonstrated this relationship between systems engineering and program success using anonymous surveys. Other prior works also provide anecdotal quantifications based on limited source data or limited conditions. There is a summary of prior works in (Honour 2004).

In the Systems Engineering Effectiveness report (Elm et al. 2007), more extensive surveys provided further detail into the relationships between specific systems engineering activities and program success. The results indicated moderately strong correlation with success for the following capabilities (in order of correlation strength): product architecture, trade studies, technical solution, integrated product teams, requirements engineering, and others. Another recent approach to quantifying the value of SE was presented in (Boehm 2008). In this work, the authors examined the existing data in the COCOMO software cost estimation model for indications about the ROI of systems engineering. There was a strong positive correlation of the "Architecture and Risk Resolution" (RESL) factor to the size of the project that indicates this SE activity.

SE-ROI Research Program

In its most recent work, the SE-ROI project (Honour 2010b) reported significant results that confirmed the 15% ratio of SE effort to total project cost through a series of detailed interviews with completed projects. Preparation for the current SE-ROI work started immediately after the publication of the author's 2004 results and has been reported in several publications. The plan for the SE-ROI research was published in (Honour 2006a) and is summarized below.

Interview Method. The SE-ROI project implemented a comprehensive and detailed gathering of information from real programs. The research obtained data from 51 programs in 16 different developing organizations. Data obtained through the interviews falls in three basic classes:

- Program success, measured in technical, cost, schedule, and subjective terms.



- Systems engineering effort, measured in labor cost against the program total cost, as a total and in each of eight categories of SE activity.
- Program characterization values (size, complexity, quality) that parameterize the expected correlation of systems engineering effort with program success.

Standardization of the data required using a structured interview process so that the interviewer could perform a consistent interpretation of the native program data into common definitions. The interviewer needed to be a senior individual with extensive program management and systems engineering experience, unbiased, and capable of probing beyond the initial question to get at the true data. The principal investigator fulfills these qualifications and performed all interviews for consistency. Many of the interview principles were selected using the pattern established by (Valerdi 2004).

Standard forms for the interviews were important and needed to reflect the best perceived *a priori* organization of SE practices to be tested. An early step in the research, therefore, was to define and obtain broad review on an ontology sufficient to provide useful and widely-accepted categorization of systems engineering activities. The ontology and resulting categorization was reported in (Honour/Valerdi 2006). Following the development of this categorization, the project developed and tested interview data sheets to obtain the necessary data (Honour 2006b). The interview data sheets were further reviewed through conference publication and detailed peer/supervisory review.

Program Access. The research is based on real programs, for which data can be both proprietary and difficult to obtain. Data obtained from programs is obviously proprietary to the parent organizations, including key business parameters of technical success, cost, schedule and risk. Therefore, all interview data is maintained by the principal investigator in accordance with proprietary data agreements with the participating organizations. These protections were necessary to obtain access to the required data.

Program Success. The data structuring of the research interviews was designed to allow four different measures of program success:

Cost Compliance. Degree of compliance with planned cost is measured by the ratio of actual cost to planned cost.

Schedule Compliance. Degree of compliance with planned schedule is measured by the ratio of actual program duration to planned program duration.

Overall Success. The success of the product/program as perceived in the viewpoint of the stakeholders, measured by a single subjective assessment on a scale of 1-to-5 allowing 0.5 scores.

Technical Quality. The technical quality of the product system is measured by the quantified compliance with the stakeholder Key Performance Parameters (KPPs) as described in (Browning/Honour 2005). The scale used for compliance is translated linearly from the original KPP values to a scale of 0=failure; 1.0=threshold acceptance; 2.0=highest objective. All KPPs are then combined using a weighted sum. Estimation of actual performance, stakeholder threshold/objective values, and weights was done in each interview from the viewpoint of the stakeholders.

Systems Engineering Activities. In this work, SE is divided into eight major activities including mission/purpose definition (MD), requirements engineering (RE), system architecting (SA), system integration (SI), verification/validation (VV), technical analysis (TA), scope management (SM), and technical management/leadership (TM). Definitions for



the eight activities are contained in (Honour/Valerdi 2006).

Prior Results. Some primary results were published in (Honour 2010b), including demographics of the interview data, definitions of basic measures, and definitions of the SE activity categories. The primary results, shown in figures 3 through 5, demonstrated the correlation of total SE effort to the first three success measures. The correlation confirms the prior-reported target SE value of 15% of total program cost.

The trend lines on the graphs show that the interview data conforms to and extends the same trends as the 2004 data, thereby corroborating both methods. The addition of the SE-ROI data provides a much better estimation of the optimum point of SE effort, although still confirming that the optimum appears to be approximately 15% of the total program cost.

It should also be noted, however, that there is a wide variation. This is to be expected, because program success depends on many factors other than SE activity. Those other confounding factors (e.g. program management, stakeholder changes, personnel quality, etc.) cause many variations that SEE cannot control.

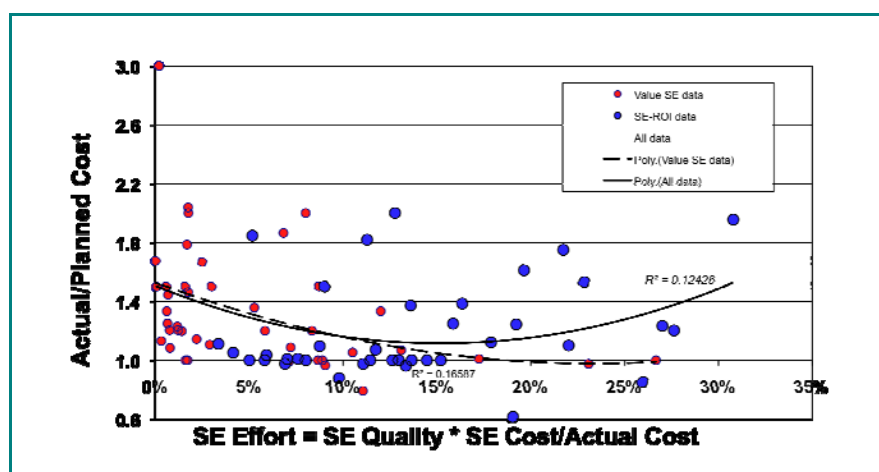


Figure 3. Correlation of SE Effort to Cost Compliance

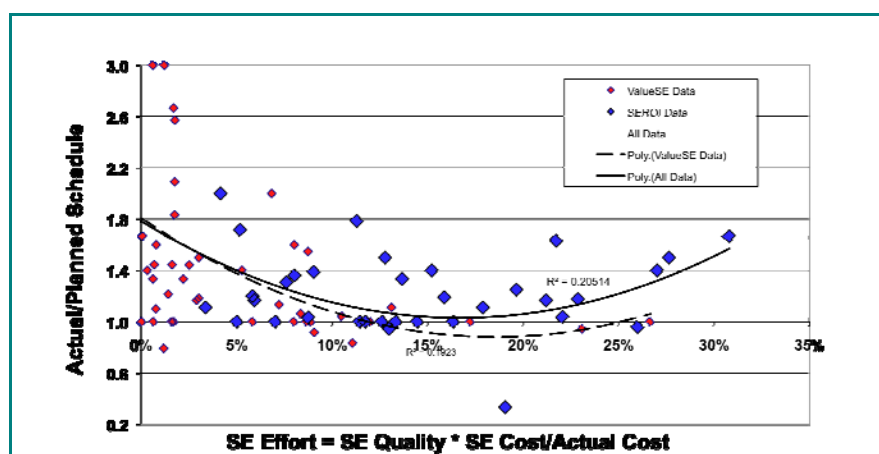


Figure 4. Correlation of SE Effort to Schedule Compliance



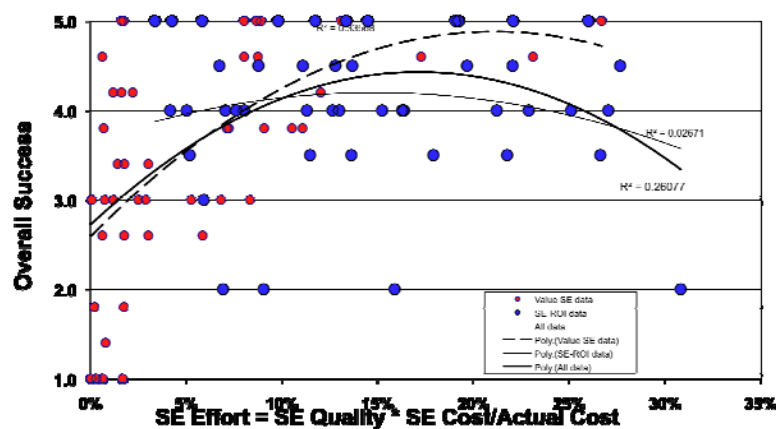


Figure 5. Correlation of SE Effort to Subjective Success

Application of Program Characterization Parameters

The SE-ROI program obtained a wealth of data about the 51 programs interviewed. In addition to the results presented using SE activities and success measures, the interviews also obtained 45 characterization parameters concerning the programs, teams, and product systems. The work presented in (Honour 2010a) correlated those characterization parameters with each other to create a useful set of 7 quantitative parameters and 7 subjective parameters that effectively characterize each program. Those parameters are shown in Figures 6 and 7. Each parameter has a specific, quantifiable definition based on the original 45 parameters and a rigorous mathematical transformation created through Principal Component Analysis.

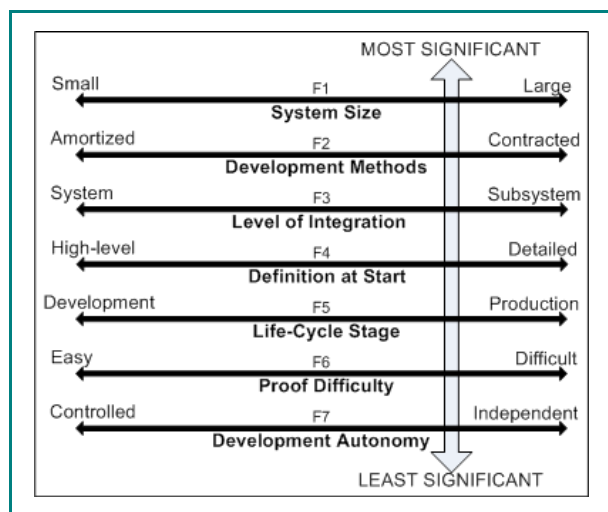


Figure 6. Quantitative Factors

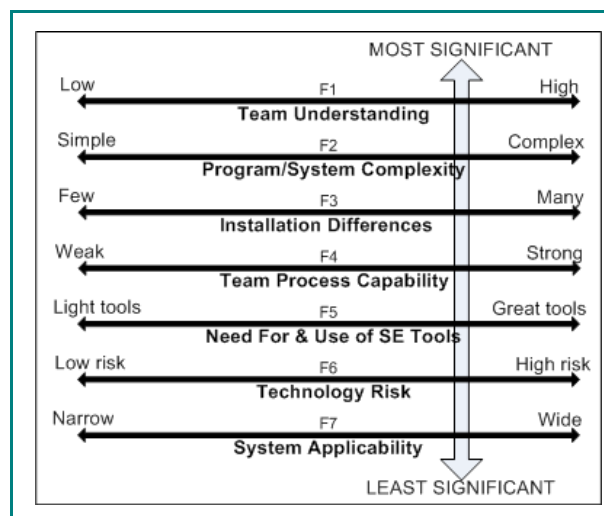


Figure 7. Subjective Factors

The typical relationship between the original characterization parameters and the useful set of parameters is shown in the left half of Figure 8, in which the System Size parameter is composed of a linear mathematical transformation from a set of the original parameters. Each of the 14 parameters of Figures 6 and 7 is made up of a similar set of original parameters, with the mathematical transformations determined by the Principal Component Analysis (PCA) described in (Honour 2010a).



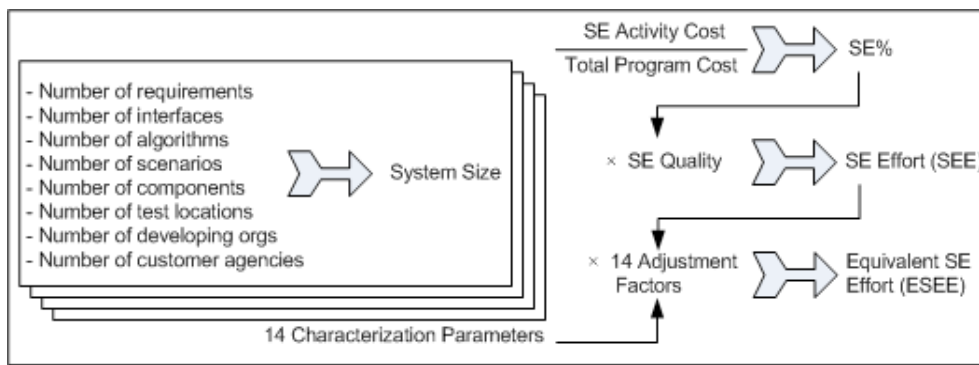


Figure 8. Mathematical Transformations of SE Effort

The prior results of (Honour 2010b), summarized in Figures 3 through 5, exhibit a wide variability due to the many confounding factors in any system development effort. SE is not the only factor that affects the success of a program. The intent of the program characterization parameters is to reduce the level of the confounding factors by adjusting the SE effort to take into account those factors. The remainder of Figure 8 shows how these characterization parameters have now been applied to modify the level of SE effort.

The underlying theory of the transformation is that each characterization parameter represents program issues that would normally require the application of more or less SE effort (SEE). It can be expected, for instance, that a larger system uses less SE effort as a proportion of the total project. The graphs of Figures 3 through 5 do not take this into account, treating all programs as equivalent size. To normalize the graphs for System Size, therefore, the SE effort applied to larger systems should be increased and that applied to smaller systems should be decreased. (e.g. For all other factors equal, perhaps a small system might require 14% SEE while a large system requires 12% SEE. If both programs actually expended 12% SEE, then the small program effectively expended less than it should have. To normalize the correlations with success, the small program data should be reduced.)

The data allows the amount of correction to be discovered by optimizing for the correlation with success as explored in (Honour 2011). The adjustment factor for each characterization parameter can be applied as a multiplicative adjustment to the SEE of each program:

$$ESEE = SEE * \prod_{j=1...14} \left(\frac{PP_j}{.5} \right)^{\frac{-Weight_j}{100}} \quad \text{Eq.1}$$

in which

- The same adjustment is applied separately to all programs based on their individual characterization parameters.
- The index j applies to the 14 characterization parameters, so that the adjustment is made simultaneously using all characterization parameters.
- PP is the percentile point (0.01 through 0.99) of the given program's characterization parameter among the same characterization parameter of all programs. Using percentile points linearizes each characterization parameter across the wide variability of program data and effectively represents the linear transformation of the PCA. (Programs that were outliers in the upper or lower 5% of percentile point were removed from the calculation.)



- The ratio allows the factor either to increase ($PP > 0.5$) or decrease ($PP < 0.5$) the ESEE depending on whether the program was in the upper or lower half of percentile point.
- The Weight is the discovery parameter, determining the true effect of the characterization parameter on the adjustment. Weights were initialized at zero (i.e. no adjustment) and allowed to vary positive or negative to discover the best weight to be used for each characterization parameter.

The result of any adjustment in graphical terms on Figures 3 through 5 is to slide the program data points horizontally. Each adjustment slides all program data points, left or right to a greater or lesser extent, depending on the characteristics of the individual program. The intent of the adjustment is to discover what values of the Weights creates the best possible correlation of the graph. By inference from the theory, those values represent the best usage of the characterization parameters to remove the effects of the many confounding variables.

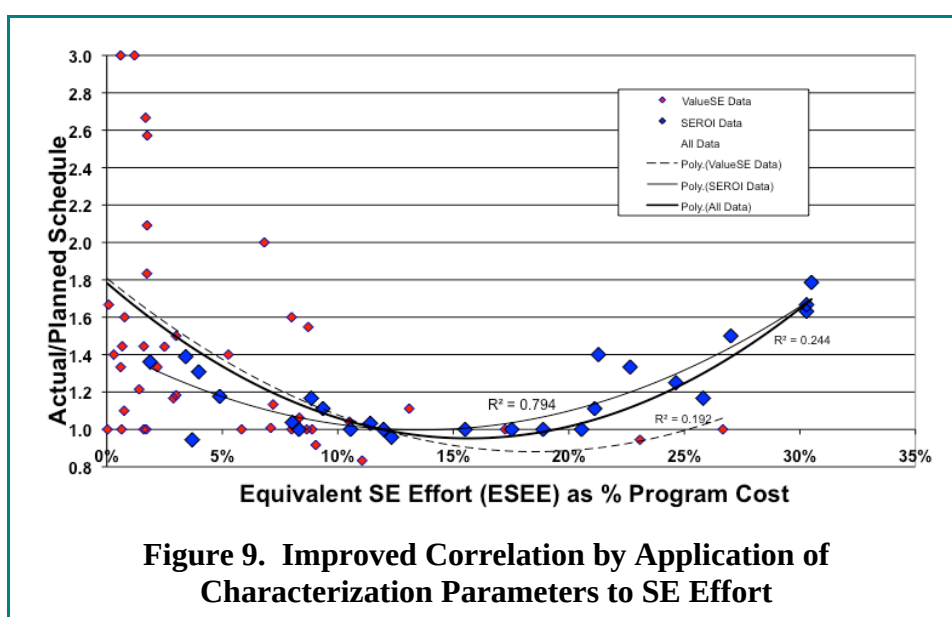


Figure 9. Improved Correlation by Application of Characterization Parameters to SE Effort

Figure 9 shows the results for one typical graph, the relationship of schedule compliance to total SE effort originally shown in Figure 4. (Only the SE-ROI data, shown as larger blue dots, was adjusted. The smaller red dots from the prior ValueSE survey work are shown for reference only.) The R^2 correlation factor for the SE-ROI data has increased from 20% to 79%, indicating the effectiveness of making the adjustments.

The adjustments made are shown in Figures 10 and 11. In each line, the increase of the parameter toward the right-hand end of its axis is accompanied by an increase or decrease in required SE effort as shown on the right.

It should be noted that the results are in some cases counter-intuitive. A more complex system, for instance, appears to require less SE as a percent of program cost, as do a system with greater installation differences, a system with greater technology risk, and a system with wider applicability. On the other hand, other results match intuition; a larger program requires less SE as a percent of the program, a program starting later required more SE to get to that point, a subsystem requires less SE than a system, and systems with greater proof difficulty and greater development autonomy require more SE. A greater exploration of the underlying interview data is indicated to further explain these results.



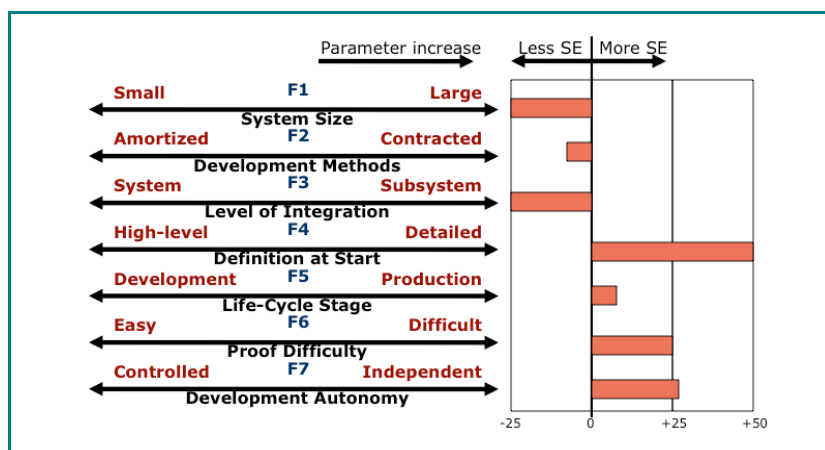


Figure 10. Adjustments Due to Quantitative Factors

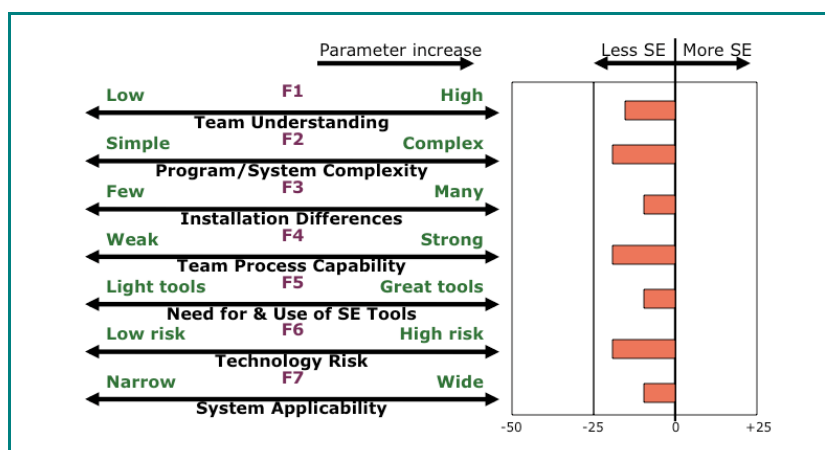


Figure 11. Adjustments Due to Subjective Factors

Nonetheless, the strong correlations indicate this major result: **Using these weighting factors, and balanced across the three success measures, the optimum level of SE effort for a median program is 15.5% of total program cost.**

A similar analysis has been performed on all eight subordinate SE activities, using the program characterizations to improve the correlation of each subordinate SE activity with the success measures. For the SE activities, in addition to the 14 characterization parameters, each activity must also be adjusted in a similar way for the confounding effects of the other 7 SE activities. Therefore, Equation 1 shown earlier is modified to index through 21 possible adjustments rather than only 14.

In all 8 SE activities, the optimum application of the characterization parameters improves the correlation factor. Figure 12 shows one example, of the improvement in the correlation of System Architecting effort (SA) with program cost compliance. In this case, the correlation improved from 13% to 62%.

Table 1 shows the optimum level of total SEE and each SE activity calculated from these correlation improvements, to balance across the four success measures.

Table 1. Optimum Levels of Equivalent Systems Engineering Activities

	MD	RE	SA	SI	VV	TA	SM	TM
Optimum Level, % of Program Cost	1.5%	2.2%	3.9%	3.0%	3.4%	2.1%	1.6%	3.5%



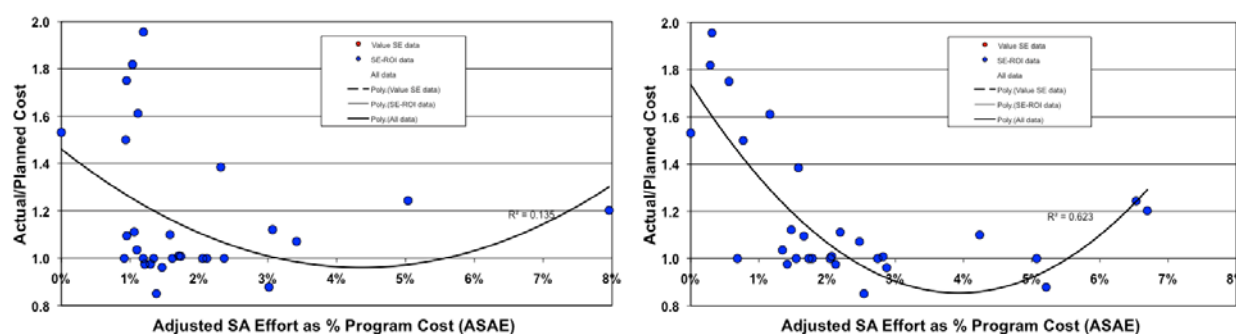


Figure 12. Correlation for System Architecting (SA) Improved from 13% to 62%.

It should be noted that the sum of the optimum values for the eight SE activities, 21.2% of program cost, does not match the optimum total SE of 15.5% as shown in Figure 9. This difference is both expected and appropriate for two reasons. First is the interaction of the eight activities; if a program optimizes the effect of any individual activity, this will de-optimize other activities. If Mission/Purpose Definition (MD), for instance, is optimized, then Requirements Engineering (RE) can likely be less. Second is the approximate nature of the statistical process used; each optimum point might vary by as much as 20-30% in actual practice.

SE Activities “Should-Be” Levels

In the prior work (Honour 2010b), the author reported the levels of SE activities that were actually used by the interviewed programs. Figure 13 shows those levels. There was a wide variation in the levels of activity used by program; for example, Technical Management/Leadership (TM) reported anywhere from near-zero to over 11% of the program total cost.

Yet these values represent only what the organizations actually expended, not what would be best. It is the purpose of this work to extend the relationships to determine the levels of each SE activity that optimize the ROI while also considering the program characterization parameters.

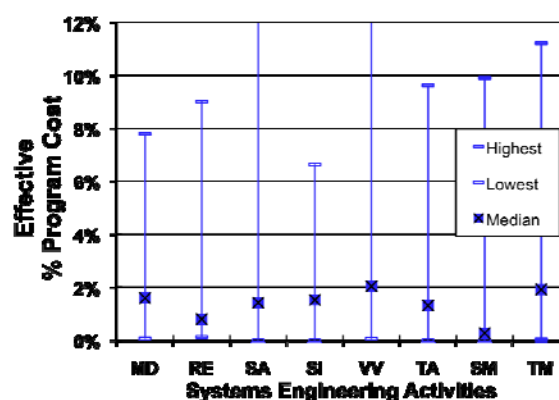


Figure 13. SE Activities Histogram

This statistical information can be used in a straightforward mathematical transformation to make an *a priori* estimation for any given program, based on the program’s characterization parameters, of the optimum value of each SE activity for best program success.

The mathematical transformation is simply the reversal of the adjustment equation 1, applied to the base optimum level:



$$OSEE = OSEE_0 * \prod_{j=1 \dots 14} \left(\frac{PP_j}{.5} \right)^{\frac{-Weight_j}{100}} \quad \text{Eq.2}$$

where

- OSEE is the optimum SEE for the program (or the optimum of each of the eight SE activities)
- OSEE₀ is the optimum level of SEE for a median program, 15.5% of program cost (or the optimum level for each SE activity as given in Table 1)
- The index j indicates each of the 14 characterization parameters (and, for the SE activities, each of the 8 SE activities),
- PP_j is the percentile point of the parameter:
 - For program parameters, it is estimated a priori based on a program estimator's evaluation of the program characterization as =0.15 for a Very Low; =0.30 for a Low; =0.50 for a Moderate; =0.70 for a High; and =0.85 for a Very High,
 - For SE activities, it is given by calculating the correction to that SE activity given by the program parameters, then applying a bracketing for ranges of Very Low, Low, Moderate, High, and Very High to use the same values as for program parameters.
- Weight_j is the weighting factor for the indexed parameter (given in Table 1 or Table 2)

This equation can be applied for total SE using OSEE₀ of 15.5% (from Figure 9), and for each of the eight SE activities (OMDE, OREE, OSAE, OSIE, OVVE, OTAE, OSME, OTME) using the data in columns of Table 1. It is then necessary to correct the optimum values for the eight activities to match the optimum for total SE by a proportional weighting.

Example Systems

System A – Shipbuilding. This program covered the development of a new ship from general specifications, an extremely large program in terms of most SE measures. Figure 14 shows the ranking of the program against the 14 derived characterization parameters. All seven of the quantitative factors were extreme, often by as much as three orders of magnitude from other programs. Such a program operates at a high degree of autonomy, using self-defined development methods similar to a commercial amortized product development. Yet the subjective parameters show a different story, that this program operated with somewhat less process definition and with only moderate use of SE tools, largely due to the team familiarity with shipbuilding development.

For each characterization parameter, the values represented in Figure 8 are translated into an estimation on the scale of Very Low (VL), Low (L), Moderate (M), High (H), or Very High (VH). These estimations are then applied to the transformation equation to determine the optimal level of effort of each SE activity and of Total SE activity, with the results in Table 2. The optimum levels are shown against the actual levels used.

The higher level (than average) of SE effort is caused by the extremely large size of the program, the high level (early) definition at start, and the greater program complexity. Mission/Purpose Definition (MD) effort is higher, also due to the early definition. Requirements Engineering (RE) effort is only slightly reduced, primarily due to team understanding. System Architecting (SA) effort is near median. System Integration (SI) effort is higher due to the proof difficulty and complexity. Verification and Validation (VV) effort and Technical Analysis (TA) effort are very high due to the difficult proof. Scope



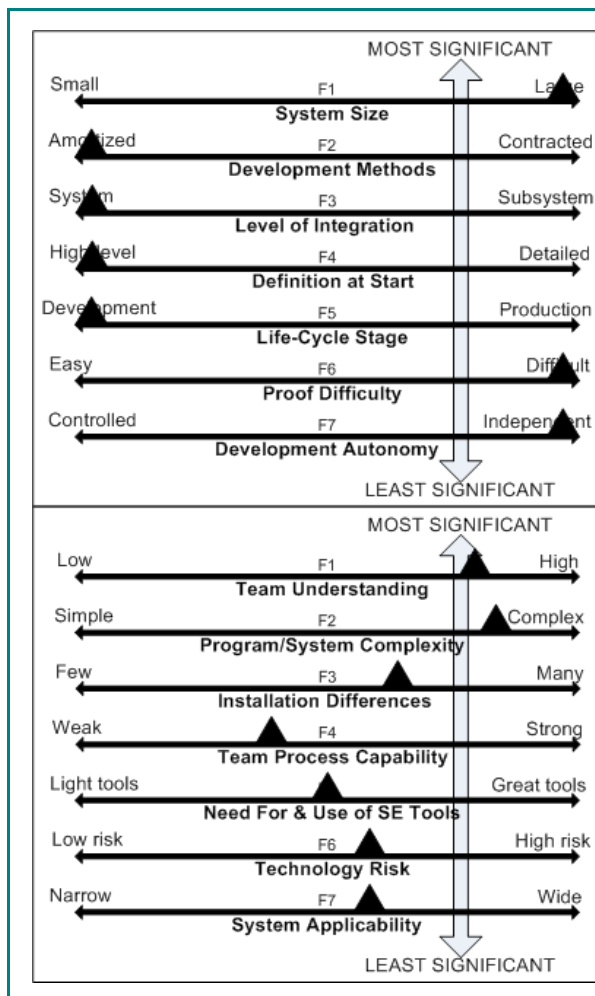


Figure 14. System A Characterization

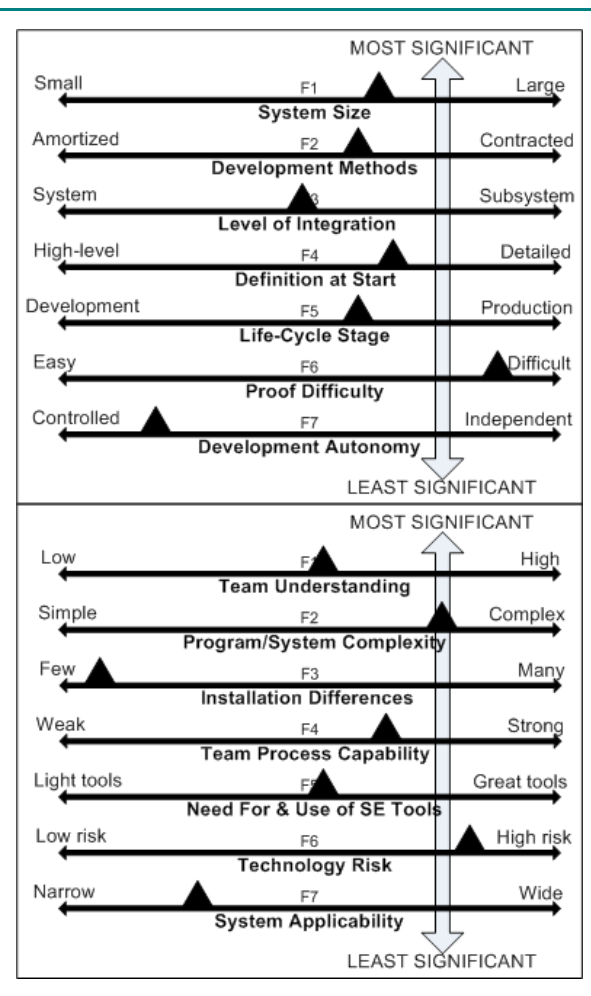


Figure 15. System B Characterization

Management (SM) effort is near median. Technical Management/Leadership effort is higher due to the complexity and size.

Table 2. System A: Optimum Levels of Equivalent Systems Engineering Activities

	MD	RE	SA	SI	VV	TA	SM	TM	Total SE
Optimum Level, % of Program Cost	1.8%	2.6%	1.6%	3.0%	4.8%	2.2%	0.6%	3.5%	20.1%
Actual Values, % of Program Cost	1.6%	1.2%	5.6%	4.2%	1.4%	1.7%	0.2%	1.4%	17.3%

This program evidenced a cost overrun of 25%, a schedule overrun of 19%, a stakeholder success rated at 2 on 1-to-5, but achieved the highest technical performance objectives. It appears that the poor cost, schedule, and stakeholder performance may be related to the variance of the actual SE effort from the optimum effort. The program spent far less than optimum in Requirements Engineering (RE), Verification & Validation (VV), Technical Analysis (TA), Scope Management (SM), and Technical Management/Leadership (TM), all of which can contribute to poorer cost/schedule/stakeholder performance. Conversely, the program spent a very large effort in System Architecting (SA) and System Integration (SI). The indication is that this program focused on “getting the system built” rather than on building the right system, and did so with insufficient technical leadership.



System B – Space system. This program covered the NASA-contracted development of a space-certified measurement system to be placed aboard the International Space Station. The program presented significant technology risk. Figure 15 shows the ranking of the program against the parameters. While the system was not particularly large, its proof was exceedingly difficult. The NASA environment led to a highly controlled development process, removing autonomy from the program. The technology and program issues led to a high program/system complexity, yet the development was only creating a single item with a single applicability.

As with System A, the values represented in Figure 15 are translated into an estimation on the scale of Very Low (VL), Low (L), Moderate (M), High (H), or Very High (VH). These estimations are then applied to the transformation equation to determine the optimal level of effort of each SE activity and of Total SE activity, with the results in Table 3.

In this case, the optimum total SE effort is near median, as are the System Architecting (SA), System Integration (SI), Scope Management (SM), and Technical Management/Leadership (TM). The Mission/Purpose Definition (MD) and Requirements Engineering (RE) are both low due to the detailed level of definition and few installation differences. The Verification & Validation (VV) and Technical Analysis (TA) efforts, however, are quite high due to the complexity and proof difficulty. The optimum levels are shown against the actual levels used.

Table 3. System B: Optimum Levels of Equivalent Systems Engineering Activities

	MD	RE	SA	SI	VV	TA	SM	TM	Total SE
Optimum Level, % of Program Cost	0.8%	2.2%	1.3%	2.3%	3.7%	2.0%	0.5%	2.7%	15.6%
Actual Values, % of Program Cost	0.9%	1.0%	2.1%	3.9%	3.6%	2.8%	0.1%	2.1%	16.5%

This program evidenced a cost overrun of 50%, a schedule overrun of 39%, a stakeholder success rated at 4.5 on 1-to-5, and scored 60% above minimal technical performance toward technical performance objectives. The stakeholder success and technical performance can be attributed to a greater effort in stakeholder-participative System Integration (SI) and Technical Analysis (SA), but these same efforts when coupled with very low Requirements Engineering (RE) likely contributed to the large cost and schedule overruns.

Conclusions

This further analysis of data from 48 interviews with system development programs reveals that the data supports a systems engineering cost estimation methodology based on optimizing the expected value of program success measures. While the prior results approximate the correlation relationship, application of the program characterization parameters significantly increases the correlation factors between systems engineering activities and program success. This increase comes about because the characterization parameters adjust the systems engineering effort based on an approximation of the many confounding factors on a project. The optimum level of SE total and for each of eight activities is shown. The types of relationships and examples of the estimation process are shown. In the final result, the SE-ROI project continues to show that it is possible and desirable to quantify the relationships between SE activities and the program success.

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Biography

Eric Honour, CSEP, INCOSE Fellow, was the 1997 INCOSE President. He has been in international leadership of the engineering of systems for over 15 years, part of a 42-year career of complex systems development and operation. He was the founding Chair of the INCOSE (International Council on Systems Engineering) Technical Board in 1994 and served as Director of the Systems Engineering Center of Excellence (SECOE). He was selected in 2000 for Who's Who in Science and Technology and in 2004 as an INCOSE Founder. He is on the editorial board for *Systems Engineering*. He has been a systems engineer, engineering manager, and program manager at Harris Information Systems, E-Systems Melpar, and Singer Link, preceded by nine years as a US Naval Officer flying P-3 aircraft. He has led or contributed to the development of 17 major systems, including the Air Combat Maneuvering Instrumentation systems, the Battle Group Passive Horizon Extension System, the National Crime Information Center, and the DDC1200 Digital Zone Control system for heating and air conditioning. Mr. Honour now heads Honourcode, Inc., a training and consulting firm offering effective methods in the development of system products. Mr. Honour has a BSSE (Systems Engineering) from the US Naval Academy, MSEE from the Naval Postgraduate School, and is a doctoral candidate at the University of South Australia based on his ground-breaking work to quantify the value of systems engineering.

